$$A = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & -1 \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Q: $\exists$ sol not all $c_i = 0$?

$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$ is in Null space of A

A is $n \times n$ For Square matrices

A is invertible

iff Null Space $= \cancel{\phi}\ \{0\}$

Computer $\rightarrow$ iff matrix rank $= n$

(dim Null + rank = # cols)

iff Cols are independent

iff rows " "

iff rows / cols span n-dim

Computer $\rightarrow$ iff $\det A \neq 0$

V_1, V_2, V_3 are linearly dependent

iff $\exists\ c_1, c_2, c_3$ not all 0

s.t. $c_1 V_1 + c_2 V_2 + c_3 V_3 = 0$

zero vector

$x - y = 2$

$y - z = 3$ redundant

$x - z = 5$

∞ sol's

or 0 sols

e.g. $x - z = 4$

$$\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

or

$$\underbrace{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}}_{= v_1} + \underbrace{\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}}_{= v_2} - \underbrace{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}}_{= v_3} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

linear dependent

v_1, v_2, v_3 Span S

iff $\forall u \in S$

$\exists c_1, c_2, c_3$ s.t. $u = c_1 v_1 + c_2 v_2 + c_3 v_3$

$$v_1 = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \quad v_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad v_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$u = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} \qquad A = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 0 & 0 \\ -1 & 1 & 0 \end{pmatrix}$$

Square

$c_1 v_1 + c_2 v_2 + c_3 v_3$

$$2c_1 - c_2 + c_3 = 7$$
$$c_1 = 8$$
$$-c_1 + c_2 = 9$$

$c_1 = 8$ $\quad -8 + c_2 = 9 \quad c_2 = 17$

$2 \cdot 8 - 17 + c_3 = 7$ $\quad \cancel{2c_1 - 8 + 17 = 7} \quad \cancel{2c_1 = -2}$

$c_3 = 8$ $\quad \cancel{c_1 = -1}$

$$V_3 = 2V_2 - 3V_1$$

$$= 2\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} - 3\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix}$$

if $c_1 V_1 + c_2 V_2 = 0$

and $c_2 \neq 0$, $c_1 \neq 0$

then $V_2 = -\dfrac{c_1}{c_2} V_1$ is obvious

$$V_1 = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} \qquad V_2 = \begin{pmatrix} -3 \\ -6 \\ -12 \end{pmatrix}$$

If ~~S~~ is linear independent $S = \{v_1, v_2, \ldots, v_n\}$

and S spans $\langle S \rangle = V$

V is a vector space

and S is a basis for V

Fact: Length of basis is independent of choice of basis

= Dimension of V

= think as free variable

Matrix Rank$\{S\}$ = dim V

$\uparrow$

a subset of S is a basis for V

size of subset = rank

$$A = \begin{array}{cc} & \text{from t} \\ \text{to u} & \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{pmatrix} \end{array}$$

$$A \cdot A^{-1} = I$$

Write System Equation

$$A \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

Solve for x, y, z

$$x + z = u$$
$$x + y + z = v$$
$$2x - y + z = w$$

$$y = v - u$$
$$= -u + v$$

$$z = u - x$$

$\cancel{x + (v - u)}$

$$2x - (v - u) + (u - x) = w$$
$$x - v + 2u = w$$
$$x = -2u + v + w$$

$$A^{-1} = \begin{array}{cc} & \text{from u} \\ \begin{array}{c} \text{to t} \\ \\ \text{to z} \end{array} & \begin{pmatrix} -2 & 1 & 1 \\ -1 & 1 & 0 \\ & & \end{pmatrix} \end{array}$$